GeoSilo

Compact geometry with explicit precision.

Install & load

```
SET allow_extension_repositories = 'allowed';
CREATE EXTENSION REPOSITORY IF NOT EXISTS query_farm WITH PREFIX 'https://duckdb-extensions.query.farm';
INSTALL geosilo FROM query_farm;  LOAD geosilo FROM query_farm;
```

01 At a glance

Store geometries as scaled integer coordinates with delta encoding. GeoSilo provides a GEOSILO column type, direct measurements and bounds, explicit encode/decode functions, and validation for untrusted BLOBs.

Native GEOMETRY → Scale + delta encode → Store / inspect / decode

02 Core SQL surface

`GEOSILO('EPSG:3857')`

Typed storage with CRS metadata and automatic scale.

`geosilo_encode(geom[,scale])`

Encode GEOMETRY to BLOB with automatic or explicit scale.

`geosilo_decode(blob)`

Decode BLOB or GEOSILO to a native GEOMETRY value.

`geosilo_metadata(blob)`

Read geometry_type, vertex_type and scale without decoding.

`ST_Area / ST_Perimeter`

Measure polygons directly, in the coordinate system's units.

`ST_XMin … ST_YMax / ST_NPoints`

Read bounds and vertex counts without decoding.

`geosilo_is_valid(blob)`

TRUE for a well-formed blob; never raises an error.

`geosilo_validate / try_geosilo_decode`

Describe the first problem, or decode malformed blobs to NULL.

03 Keep in mind

Precision follows the scale

Encoding rounds coordinates onto a grid with spacing 1/scale. A scale of 100 keeps hundredths of a coordinate unit. Projected-meter defaults use 100; degrees default to 10^7. CRS labeling does not transform coordinates.

04 Store a projected polygon

```
CREATE TABLE plots (
  geom GEOSILO('EPSG:3857')
    USING COMPRESSION zstd
);
INSERT INTO plots VALUES (
  'POLYGON((0 0,10 0,10 10,0 10,0 0))'
    ::GEOMETRY
);
```

Inserting GEOMETRY automatically encodes it for the typed column.

05 Inspect and measure directly

```
SELECT ST_Area(geom) AS area,
       ST_Perimeter(geom) AS perimeter,
       geosilo_metadata(geom) AS metadata
FROM plots;
```

Area 100; perimeter 40; integer scale 100. These are planar measurements.

06 Set a scale for a BLOB pipeline

```
WITH encoded AS (
  SELECT geosilo_encode(
    'POINT(1.234 2.345)'::GEOMETRY,100
  ) AS blob
)
SELECT geosilo_is_valid(blob) AS ok,
  geosilo_decode(blob)::VARCHAR AS wkt
FROM encoded;
```

Returns true, POINT (1.23 2.35): decoding retains the chosen quantization.

07 In practice

Spatial compatibility

Either extension can load first. Native ST_* overloads follow spatial's semantics, including MULTI* geometries and collections: areas subtract holes, polygons have length 0 and empty geometries give NULL. Other ST_* functions decode through the implicit cast.

Full docs & examples ↗ Source ↗